

Network Layer

1 Overview of the Network Layer

The network layer is responsible for delivering packets (datagrams) from a source host to a destination host across multiple routers.

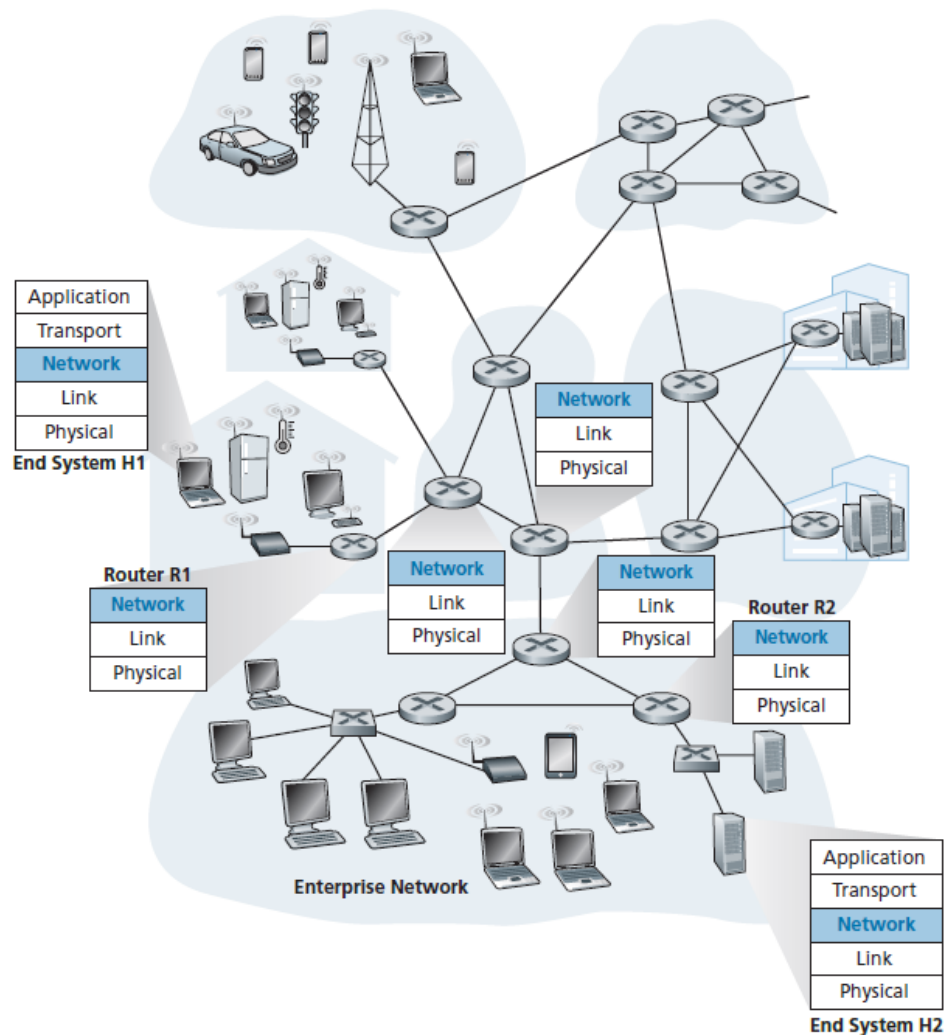


Figure 1: A simple network with hosts and routers.

Major Functions

Network Layer Responsibilities

- Encapsulating transport-layer segments into datagrams.
- Delivering datagrams to the receiver's transport layer.
- Forwarding packets through routers.
- Computing routes across networks.

Routers implement **only the network layer**, not the transport or application layers.

2 Data Plane vs. Control Plane

Two Components of the Network Layer

Data Plane: Local per-router operations (forwarding). **Control Plane:** Network-wide route computation.

2.1 Forwarding

Forwarding is the router's function of moving a packet from the input link to the appropriate output link. This process occurs at high speed (nanoseconds).

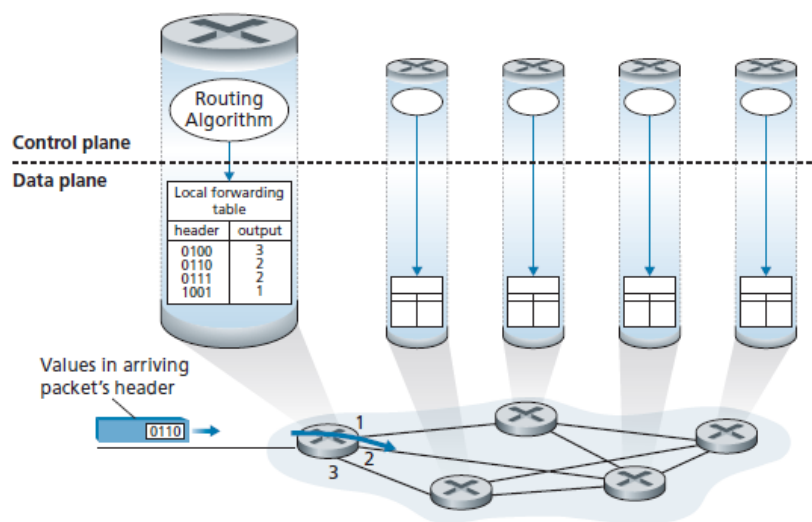


Figure 2: Example forwarding table used by a router.

2.2 Routing

Routing determines the end-to-end path from source to destination. Unlike forwarding, routing takes longer (seconds) and is implemented in software.

- Routers exchange routing messages.
- Routing protocols compute forwarding tables.

2.3 Driving Analogy

- **Forwarding** = Choosing an exit in a single interchange.
- **Routing** = Planning the complete trip across cities or states.

3 Control Plane Approaches

3.1 1. Traditional Per-Router Control

Each router runs routing algorithms (e.g., OSPF, RIP) and exchanges routing messages. Each router builds its **own forwarding table**.

3.2 2. Software-Defined Networking (SDN)

SDN separates the control and data planes. A centralized controller computes forwarding tables and distributes rules to routers.

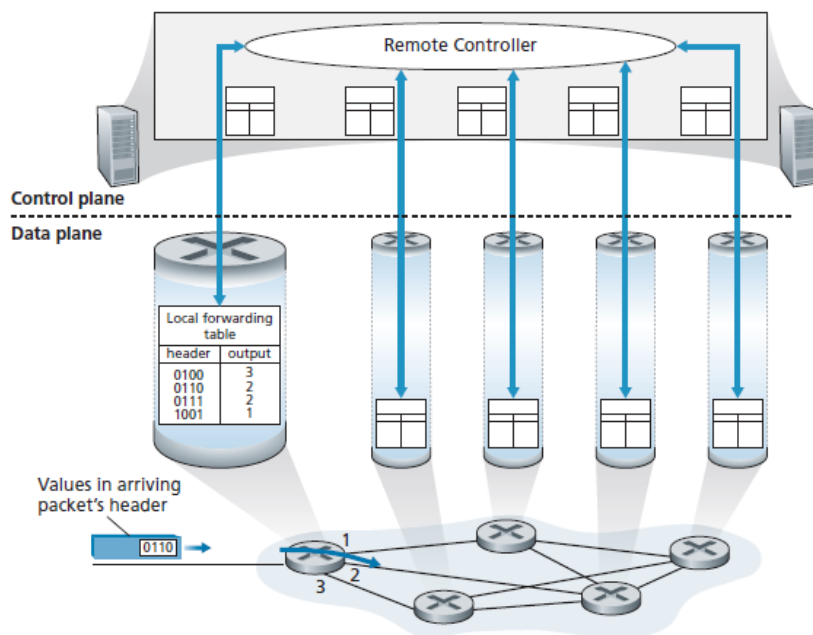


Figure 3: Centralized SDN controller communicating with routers.

SDN Key Idea

Routers perform only fast data-plane forwarding, while the controller handles all routing intelligence.

4 Network Service Model

The network service model defines the “type of service” provided by the network to the applications.

Possible Network-Layer Services

- Guaranteed delivery
- Bounded delay
- In-order delivery
- Guaranteed minimum bandwidth
- Security (integrity, encryption, authentication)

4.1 Internet Service Model: Best Effort

The Internet supports only **best-effort delivery**:

- No delivery guarantee
- No delay guarantee
- No bandwidth guarantee
- No order guarantee

Despite this, adaptive protocols (e.g., TCP, DASH streaming) make the Internet robust.